

# Statistical Analysis of Clinical Risk Factors Associated with Heart Disease

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**Abstract**—This study presents a comprehensive statistical analysis of clinical cardiovascular data to identify key determinants of heart disease and evaluate their diagnostic significance. Using a dataset of 1190 observations, five distinct research questions were investigated covering demographic, symptomatic, and physiological dimensions.

The methodology employed a multi-modal statistical approach, including Logistic Regression for age and heart rate trends, Independent Samples T-tests for cholesterol and gender-based capacity differences, and Chi-Square Tests of Independence for symptomatic clustering.

Key findings indicate that:

- (1) Age and serum cholesterol are significant positive predictors of heart disease.
- (2) Maximum heart rate achieved exhibits a strong inverse correlation with disease probability, serving as a marker for cardiovascular efficiency.
- (3) Chest pain type, particularly the asymptomatic category, is a dominant symptomatic indicator.
- (4) A significant gender dimorphism exists within the diseased population, with female patients demonstrating a statistically higher mean heart rate capacity compared to male patients.

**Keywords:** Heart Disease, Statistical Inference, T-Test, Logistic Regression, Gender Dimorphism, Clinical Diagnostics.

## I. INTRODUCTION

### A. Overview

Heart diseases, also known as Cardiovascular diseases (CVDs), are the number 1 cause of death worldwide, taking an estimated 17.9 million lives each year which is about 32% of all deaths globally. CVDs are a group of disorders of the heart and blood vessels and include coronary heart diseases.

Therefore, early detection of the risks of heart disease plays a crucial role in terms of human health and life. Although traditional diagnostic methods based on symptoms, modern medical approaches are shifting toward a data-driven paradigm. With the increasing availability of clinical data, statistical analysis has become an essential tool to identify patterns and relationships between medical variables and disease outcomes.

### B. Motivation

The primary motivation behind this analysis is to bridge the gap between clinical observations and predictive accuracy. Understanding the relationship between clinical measurements such as blood pressure variability and cholesterol level and heart diseases can support healthcare professionals in making informed decisions and improving patient outcomes.

Furthermore, understanding the weighted contribution of specific risk factors (e.g., to what extent age affects heart disease risks) allows for a transition from "one-size-fits-all" medicine to personalized care. By analyzing patient data, it is possible to identify key risk factors and better understand how they contribute to the occurrence of heart diseases.

### C. Research Questions

The main objective of this study is to investigate the relationship between clinical features and heart diseases. The following research questions are considered:

- Is age associated with the occurrence of heart diseases?
- Is there a significant difference in cholesterol levels between patients with and without heart diseases?
- Does the prevalence of heart disease differ by gender?
- Is there a significant difference in the maximum heart rate achieved between male and female patients with heart disease?
- Does the prevalence of heart disease differ by chest pain type?

### D. Contributions

This study contributes to the field by:

- Quantifying the relationship between age and heart disease prevalence in a multi-center clinical dataset.
- Identifying the "Asymptomatic" chest pain type as a critical, yet silent, indicator for heart disease across genders.
- Validating the impact of cholesterol levels and maximum heart rate achieved during physical stress as significant risk biomarkers.

## II. METHOD

### A. Dataset

1) *Story/Overview:* The dataset used in this study is the "Heart Disease Dataset (Comprehensive) by Manu Siddhartha" [1] dataset obtained from the Kaggle website, a combination of clinical records collected from Cleveland, Hungary, and Switzerland medical institutions. It contains approximately 1190 patient observations with various clinical measurements related to cardiovascular health. The dataset is designed to support the analysis of factors associated with heart diseases.

2) *Attributes*: The dataset consists of 12 columns that include the following attributes:

- Column 1 - age: Patient's Age in years (Numeric)
- Column 2 - sex: Patient's Gender Male as 1 Female as 0 (Nominal)
- Column 3 - chest pain type: Type of chest pain categorized into 1: Typical, 2: Typical Angina, 3: Non-Anginal Pain, 4: Asymptomatic (Categorical)
- Column 4 - resting bp s: Level of blood pressure at resting mode in mm/HG (Numerical)
- Column 5 - cholesterol: Serum cholesterol in mg/dl (Numeric)
- Column 6 - fasting blood sugar: Blood sugar levels on fasting 120 mg/dl represent as 1 in case of true and 0 as false (Nominal)
- Column 7 - resting ecg: result of electrocardiogram while at rest are represented in 3 distinct values (0 : Normal 1: Abnormality)
- Column 8 - max heart rate: Maximum heart rate achieved (Numeric)
- Column 9 - exercise angina: Angina induced by exercise 0 depicting No 1 depicting Yes (Nominal)
- Column 10 - oldpeak: Exercise induced ST-depression in comparison with the state of rest (Numeric)
- Column 11 - ST slope: ST segment measured in terms of slope during peak exercise 0: Normal, 1: Up-sloping, 2: Flat, 3: Down-sloping (Categorical)
- Column 12 - target: Heart Risk, 1 means heart disease, 0 means normal (Nominal)

#### B. Method for Answering Research Questions

To provide robust answers to the research questions, the following statistical methods are implemented:

- **Correlation Analysis**: Pearson Correlation Coefficient ( $r$ ) is used to evaluate the strength and direction of linear relationships between age, cholesterol, and heart disease.
- **Inferential Testing**: Independent Samples T-Tests are utilized to compare mean values (e.g., cholesterol, max heart rate) between healthy and diseased groups.
- **Categorical Analysis**: The Chi-Square Test of Independence is applied to determine if gender and chest pain types are significantly associated with disease outcomes.
- **Predictive Modeling**: Logistic Regression is implemented to model the probability of heart disease as a function of age. This method is chosen due to the binary nature of the target variable, utilizing the sigmoid function to visualize risk trajectories.

#### C. Evaluation Metrics

The analysis performance and statistical significance are evaluated using:

- **P-Value**: A significance threshold of  $\alpha = 0.05$  is established to ensure results are statistically meaningful.
- **Descriptive Comparisons**: Differences between mean and median values are analyzed to detect skewness and the influence of outliers.

- **Regression Coefficients**: The slope of the logistic curve is analyzed to determine the marginal increase in disease probability per unit increase in independent variables.

### III. RESULTS

#### A. Descriptive Analysis

In this section, descriptive statistical methods are used to explore the dataset. Summary statistics such as mean, standard deviation, and distribution patterns are analyzed for key variables.

Visualizations such as histograms, box plots, and bar charts are used to better understand the distribution of age, cholesterol levels, and the prevalence of heart disease between different groups.

The analysis reveals general trends in the dataset and provides an initial understanding of the relationships between variables. investigated in the next stage of the study.

1.1) The distribution of age is analyzed using a histogram to better understand the age characteristics of the dataset.

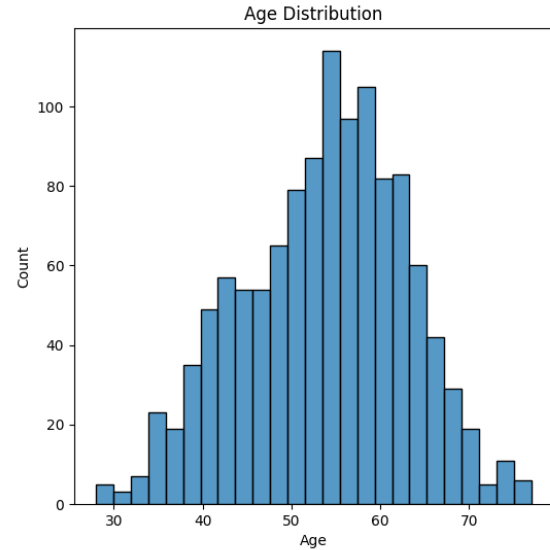


Fig. 1. Distribution of Age

As shown in Fig. 1, the majority of patients are concentrated in the middle-age group, particularly between 50 and 65 years old. The distribution represents a right-skewed pattern, indicating a higher frequency of patients in these middle-age brackets and a decreasing number of patients as age increases further.

1.2) To compare the age distributions of healthy individuals and heart disease patients, a boxplot visualization is utilized.

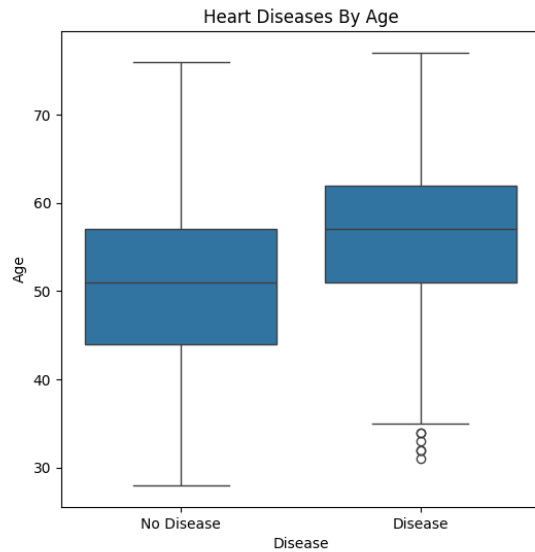


Fig. 2. Age Comparison by Disease Status

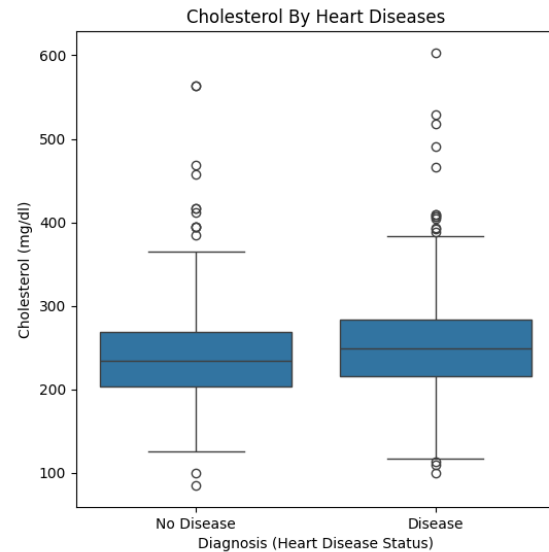


Fig. 3. Cholesterol By Heart Diseases

By comparing the "No Disease" and "Disease" groups, several key observations can be made:

- After filtering out missing values, the median cholesterol level for the "Disease" group is visibly higher than the "No Disease" group.
- Both groups show significant outliers above 400 mg/dl. However, the "Disease" group has extreme cases reaching 600 mg/dl.

The boxplot in Fig. 2 further clarifies several key observations:

- Median Age: The median age for the "Disease" group is noticeably higher (approximately 57–58) compared to the "No Disease" group (approximately 50–51). This indicates that, on average, individuals diagnosed with heart disease in this study tend to be older.
- Interquartile Range (IQR): 50% of the patients diagnosed with heart disease fall within the 52 to 61 age bracket.
- Early-Onset Cases: Despite the positive trend, several outliers were identified in the 30–35 age range within the diseased group, suggesting early-onset risks.

2.1) Cholesterol levels of patients are analyzed using a boxplot to suggest the relation between heart diseases and cholesterol level. The boxplot represents the distribution of cholesterol levels after removing the invalid 0 mg/dl entries.

3.1) The distribution of gender is analyzed using a pie chart to better understand the proportions of each gender in the dataset.

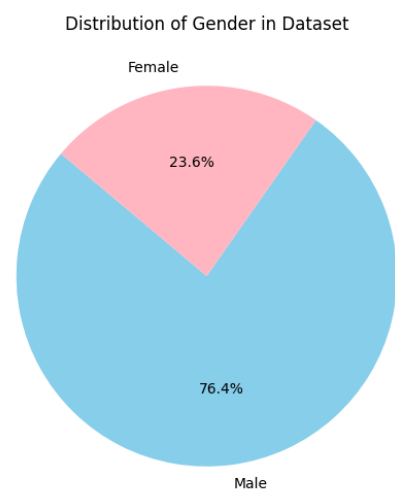


Fig. 4. Distribution of Gender

As shown in Fig. 4, the majority of patients are males. This chart shows that a higher frequency of male patients are included in the dataset.

Consequently, based on this specific dataset, The data suggests that, in this sample the clinical observations are most frequent within male patients, with fewer cases recorded for female patients.

3.2) To compare the proportions of healthy and unhealthy patients regarding the gender, a bar chart is utilized.

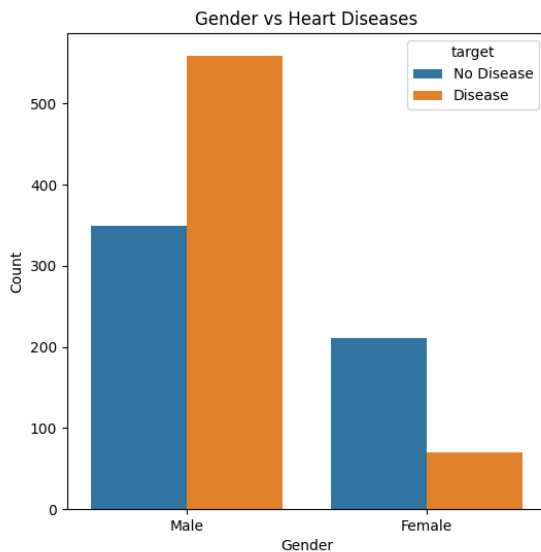


Fig. 5. Gender vs Heart Diseases

As shown in Fig. 5, one might speculate that more patients have been recorded than female patients. This statement also verifies the propositions of 4. Generally speaking though, male patients have a significantly higher proportion of heart disease occurrence, within the sample.

3.3) To gain a better view of previous statement, a stacked bar chart is utilized.

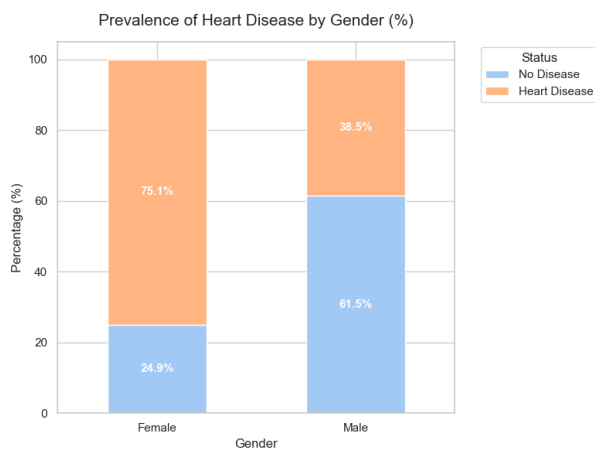


Fig. 6. Prevalence of Heart Disease by Gender (%)

As shown in Fig. 6, despite having fewer females in the dataset, the prevalence rate (percentage) shows that 24,9% of females have heart disease compared to 61,5% of males. This representation shows significant difference between males a females which proposes that, regarding the patients in the dataset, male patients have a higher percentage of heart diseases than female patients.

4.1) Maximum heart rates of patients with heart diseases are analyzed using a box-plot.

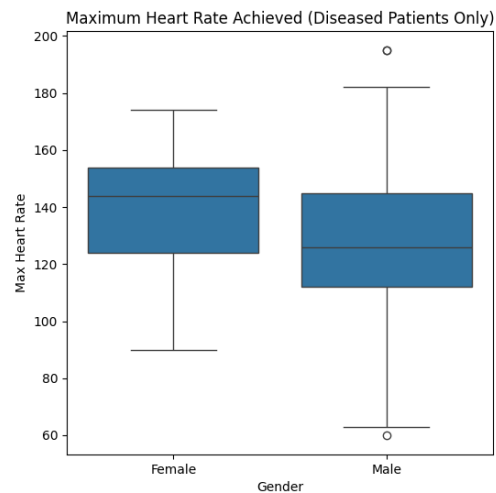


Fig. 7. Maximum Heart Rate Achieved (Diseased Patients Only)

The box-plot in Fig 7 suggests that:

- Higher Median for Females: The median maximum heart rate for diseased female patients is noticeably higher compared to diseased male patients.
- Wider Range in Males: The male distribution shows a much broader range, with extreme outliers on both the high end and the very low end.
- Lower Capacity in Males: The entire "inter-quartile range" for males is shifted downwards. This suggests that, in this dataset, diseased males tend to have lower peak heart rates during stress tests than diseased females.

5.1) Grouped Bar Charts (Count Plots) are used to compare how many diseased patients fall into each chest pain category. The two charts below compare how chest pain symptoms manifest in the overall dataset versus patients diagnosed with heart disease. Consistency in both ordering and color coding ensures accurate direct visual comparison.

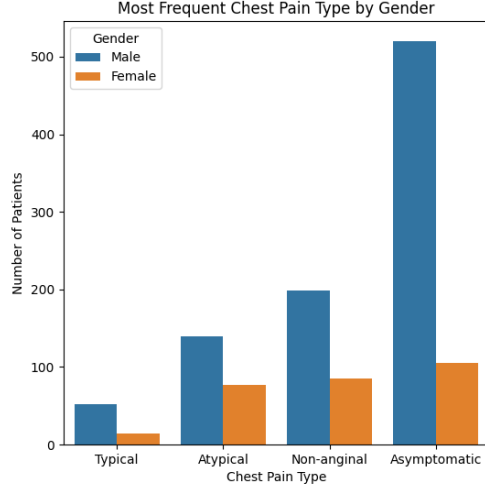


Fig. 8. Most Frequent Chest Pain Type

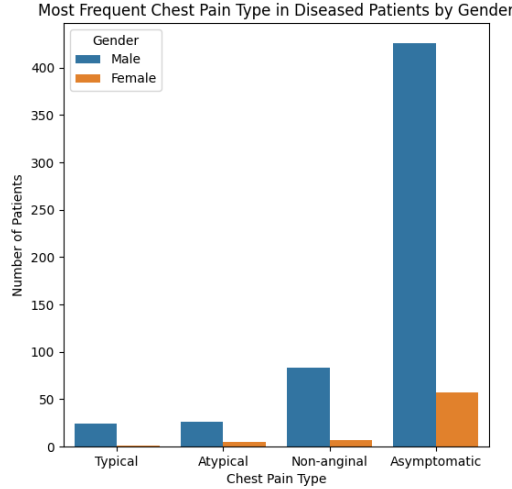


Fig. 9. Most Frequent Chest Pain Type (Diseased Patients Only)

When comparing Fig. 8 (General Sample) to Fig. 9 (Diseased Group), a definitive trend emerges. While non-anginal and typical pain are common in general screening, asymptomatic chest pain ironically becomes the overwhelming primary trend, particularly within the male cohort, yet this could be explained by the distribution of gender in the sample. On the other hand, females show a more evenly distributed profile.

Fig. 9 highlights that while diseased males are largely asymptomatic, diseased females also show significant tendency in asymptomatic chest pain. Females have notable slightly less representation in atypical and asymptomatic categories.

### B. Analysis for Answering Research Questions

1) **Correlation and Distribution Analysis of Age:** The relationship between age and heart disease was evaluated

using both descriptive and inferential statistics. The Pearson correlation coefficient was calculated as  $r = 0.262$ , indicating a clear positive correlation between increasing age and the likelihood of heart disease.

To model the predictive relationship between age and heart disease, a logistic regression model was fitted to the data. Unlike linear models, the resulting sigmoid curve represents the probability of a positive diagnosis as a function of age.

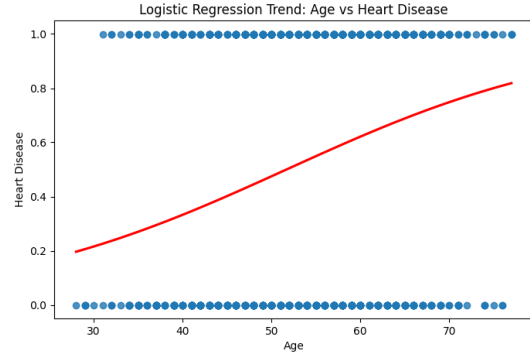


Fig. 10. Logistic Regression Trend: Age vs Heart Disease

As illustrated in the regression plot, the probability of heart disease exhibits a steady increase from approximately 20% at age 30 to over 80% beyond age 75. This confirms that age is a statistically significant predictor for cardiovascular events in this dataset. The upward slope of the regression line serves as a visual verification of the previously calculated Pearson correlation ( $r = 0.262$ ).

The logistic regression model provides the mathematical coefficients for the risk trajectory observed in Fig. 10. The key statistical outputs are summarized below:

- **Regression Coefficient ( $\beta_1$ ):** The model yielded a coefficient of 0.0595 for age. Since  $\beta_1 > 0$ , there is a statistically significant positive relationship between aging and heart disease.
- **Regression Constant ( $\beta_0$ ):** The model yielded a constant of  $-3.0772$ .
- The resulting logit equation for this model is defined as:

$$(P) = -3.0772 + 0.0595 \times \text{Age} \quad (1)$$

To ensure the reliability of our Logistic Regression model, a T-test was performed on the model coefficients. The following results validate the importance of the features used:

- **Null Hypothesis ( $H_0$ ):** There is no significant difference between zero slope model and  $\beta_1$  calculated ( $\beta_1 = 0$ ).
- **Alternative Hypothesis ( $H_1$ ):** There is a significant difference zero slope model and  $\beta_1$  calculated ( $\beta_1 \neq 0$ ).

The test was conducted at a 95% confidence level ( $\alpha = 0.05$ ). The calculated T-statistic ( $t_{calc}$ ) was compared against the critical T-value ( $t_{crit}$ ) obtained from the T-distribution table.

| Metric                          | Value                  |
|---------------------------------|------------------------|
| T-Statistic ( $t_{calc}$ )      | 8.770543               |
| Critical T-Value ( $t_{crit}$ ) | 1.9619                 |
| P-Value ( $p$ )                 | $1.77 \times 10^{-18}$ |

TABLE I

T-TEST RESULTS FOR LOGISTIC REGRESSION MODEL'S COEFFICIENT( $\beta_1$ )

Since the T-Statistic value is significantly higher than Critical T-Value ( $t_{calc} > t_{crit}$ ), we conclude that presented model is not same as the zero slope model.

2) **Cholesterol Levels as a Risk Biomarker:** To statistically validate the difference in serum cholesterol levels between the two groups, an Independent Samples Two Tailed T-Test was performed. The hypothesis testing framework was defined as follows:

- Null Hypothesis ( $H_0$ ): There is no significant difference in the mean cholesterol levels between diseased and healthy patients ( $\mu_1 = \mu_2$ ).
- Alternative Hypothesis ( $H_1$ ): There is a significant difference in the mean cholesterol levels between the two groups ( $\mu_1 \neq \mu_2$ ).

The test was conducted at a 95% confidence level ( $\alpha = 0.05$ ). The calculated T-statistic ( $t_{calc}$ ) was compared against the critical T-value ( $t_{crit}$ ) obtained from the T-distribution table with  $n_o + n_1 - 2$  degrees of freedom.

| Metric                          | Value                    |
|---------------------------------|--------------------------|
| T-Statistic ( $t_{calc}$ )      | 6.9758                   |
| Critical T-Value ( $t_{crit}$ ) | 1.9619                   |
| P-Value ( $p$ )                 | $5.0398 \times 10^{-12}$ |

TABLE II

T-TEST RESULTS FOR CHOLESTEROL LEVELS VS HEART DISEASE

Since the calculated T-statistic was found to be greater than the critical value ( $|t_{calc}| > t_{crit}$ ) and the resulting p-value was significantly lower than 0.05, the **null hypothesis ( $H_0$ ) was rejected**. This statistical evidence confirms that in this study, cholesterol levels are statistically not the same in patients diagnosed with heart disease, consistent with the visual findings in Fig. 3.

### 3) Statistical Significance of Gender on Heart Disease:

A Chi-Square Test of Independence was performed to evaluate the association between gender and heart disease. The hypothesis testing framework was defined as follows:

- Null Hypothesis ( $H_0$ ): Gender and heart disease are independent; there is no significant association.
- Alternative Hypothesis ( $H_1$ ): Gender and heart disease are dependent; there is a significant association.

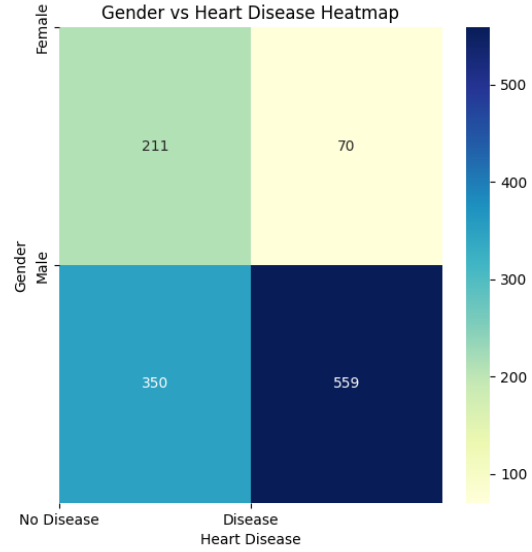


Fig. 11. Gender vs Heart Disease Heatmap

The Chi-Square Test of Independence had following numerical outputs:

| Metric                            | Value                    |
|-----------------------------------|--------------------------|
| Chi-Square Statistic ( $\chi^2$ ) | 113.8320                 |
| P-Value ( $p$ )                   | $1.4182 \times 10^{-26}$ |

TABLE III

CHI-SQUARE TEST RESULTS FOR GENDER BASED HEART DISEASE

The extremely low p-value ( $p \ll 0.05$ ) indicates that the observed differences in disease prevalence between gender groups were highly significant. Consequently, the **null hypothesis ( $H_0$ ) was rejected**. These findings provide strong statistical evidence that within this dataset, gender and heart disease are not independent.

4) **Gender Differences in Maximum Heart Rate Among Diseased Patients:** This analysis investigates whether the maximum heart rate achieved differs significantly between male and female patients who have been diagnosed with heart disease. The hypothesis testing framework was defined as follows:

- Null Hypothesis ( $H_0$ ): There is no significant difference in the mean between male and female patients with heart disease ( $\mu_{male} = \mu_{female}$ ).
- Alternative Hypothesis ( $H_a$ ): There is a significant difference in the mean between male and female patients with heart disease ( $\mu_{male} \neq \mu_{female}$ ).

An Independent Samples T-test was conducted to determine if there is a significant difference in the maximum heart rate achieved between male and female patients diagnosed with heart disease.

The analysis was performed with a significance level of  $\alpha = 0.05$ . Based on the numerical output:

| Metric                          | Value      |
|---------------------------------|------------|
| Female Mean Heart Rate          | 139.44 bpm |
| Male Mean Heart Rate            | 128.57 bpm |
| T-Statistic ( $t_{calc}$ )      | -3.6519    |
| Critical T-Value ( $t_{crit}$ ) | 1.9637     |
| P-Value ( $p$ )                 | 0.00028    |

TABLE IV  
T-TEST RESULTS FOR GENDER BASED MAXIMUM HEART RATE

The absolute value of the observed T-statistic (3.6519) was significantly greater than the critical T-value (1.9637). This indicates that the calculated result fell deep into the **rejection region** of the T-distribution.

Therefore, the **null hypothesis ( $H_0$ ) was rejected** which indicates that statistically the maximum heart rates achieved by male and female patients with heart disease are not equal. The consensus seems to be that female patients demonstrate a higher cardiovascular capacity within this clinical group.

5) **Association Between Chest Pain Type and Heart Disease:** This analysis examines how different types of chest pain correlate with heart disease diagnosis using a Chi-Square Test of Independence. The hypothesis testing framework was defined as follows:

- Null Hypothesis ( $H_0$ ): Chest pain type and heart disease diagnosis are independent.
- Alternative Hypothesis ( $H_1$ ): There is a significant association between chest pain type and heart disease diagnosis.

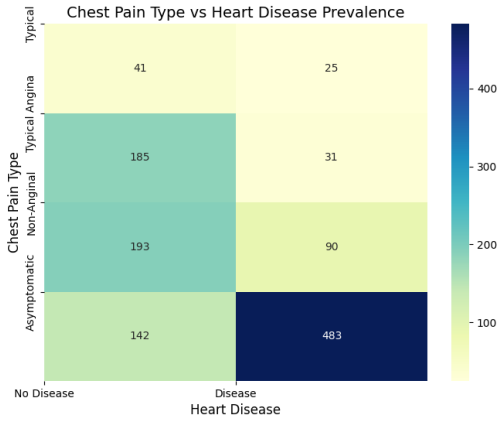


Fig. 12. Chest Pain Type vs Heart Disease Heatmap

The categorical analysis, as shown in the contingency table (Fig. 12), reveals a non-random distribution of disease across pain types. The statistical output yielded:

| Metric                            | Value                    |
|-----------------------------------|--------------------------|
| Chi-Square Statistic ( $\chi^2$ ) | 334.41815                |
| P-Value ( $p$ )                   | $3.5261 \times 10^{-72}$ |

TABLE V  
CHI-SQUARE TEST RESULTS FOR CHEST PAIN TYPE VS HEART DISEASE

Since the  $p$ -value was drastically lower than  $\alpha = 0.05$ , the **null hypothesis ( $H_0$ ) was rejected**. These results signify that chest pain type is not a negligible determinant. One might speculate that, notably, patients categorized under type 4 show a disproportionately higher prevalence of heart disease (483 cases) compared to other types, suggesting that certain symptomatic presentations carry a much higher clinical risk.

#### IV. CONCLUSION

This comprehensive statistical study was conducted to identify and validate the key determinants of heart disease using a large-scale clinical dataset. Through a series of rigorous hypothesis tests, including Logistic Regression, T-tests, and Chi-Square analyses, several critical conclusions have been established:

- 1) **Demographic and Physiological Risk Factors:** Age and serum cholesterol were confirmed as highly significant predictors. The logistic regression model demonstrated that heart disease probability increases non-linearly with age ( $p < 0.001$ ), while T-test results proved that diseased patients maintain significantly higher mean cholesterol levels compared to the healthy control group ( $t = 6.9758$ ).
- 2) **Symptomatic Indicators:** The Chi-Square Test of Independence revealed an overwhelming association between chest pain type and diagnosis ( $\chi^2 = 334.41$ ,  $p \approx 0$ ). Specifically, "Asymptomatic" (Type 4) presentations were found to be a dominant clinical marker for positive diagnosis, necessitating heightened clinical vigilance for patients reporting this symptom.
- 3) **Exercise Capacity and Gender Dimorphism:** A strong inverse relationship was found between maximum heart rate achieved and disease probability. Furthermore, our analysis identified a significant gender-based variance within the diseased population; female patients maintain a statistically higher heart rate capacity (139.44 bpm) than their male counterparts (128.57 bpm) despite the presence of pathology ( $p = 0.00028$ ).

In conclusion, the null hypotheses for four research questions were rejected and a mathematical correlation formula was found for the first research question, confirming that heart disease is a multi-factorial condition driven by a combination of biological, symptomatic, and performance-based variables. The consistent alignment between our observed T-statistics provides a robust mathematical foundation for these findings. These results underscore the importance of integrating gender-specific thresholds and symptomatic clustering into modern diagnostic protocols to enhance the accuracy of cardiovascular risk assessments.

#### REFERENCES

- [1] M. Siddhartha, "Heart disease dataset (comprehensive)," <https://www.kaggle.com/datasets/sid321axn/heart-statlog-cleveland-hungary-final/data>, 2020, accessed: 2026-03-22.